



Bench Validation Toolkit

Scope • DMM • Logic analyzer • Current probe • Thermal camera — what each does

Your tools determine how fast you can debug

Audience: PCB engineers building their bench

Tool selection • Use cases • Budget options • Setup



Bench Tier System

Hobby tier — ₹15-30k

Basic scope + DMM
+ soldering iron.

Pro tier — ₹1-3 L

Mixed-signal scope +
logic analyzer + current probe.

Lab tier — ₹5-10 L

High-BW scope + signal gen +
spectrum analyzer.

Production — ₹10 L+

Bed-of-nails fixture +
FCT station + chamber.

Cloud tier

Use shared lab equipment
(makerspace, university).

Field tier

Portable scope + DMM
for on-site debug.

Oscilloscope — Specs to Know



Spec	Hobby tier	Pro tier	Notes
Bandwidth	100 MHz	500 MHz - 1 GHz	5× max signal freq
Channels	2	4	More = more signals at once
Sample rate	1 GS/s	5-10 GS/s	4× bandwidth typical
Memory depth	1 MS	100+ MS	Longer = capture more time
Probes	1× std + 10× std	Differential + active	High-Z probes for high-Z circuits
Display	7" LCD	10-12" + HD	Larger = better visibility
Trigger	Edge, pulse	Serial, video, math	More = better debug

Recommended models

Hobby: Rigol DS1054Z (₹35k) or Siglent SDS1104X-E (₹40k)
Pro: Keysight DSOX2024A or Tek MSO22
Ultra-budget: Hantek 6022BE (USB, ₹4k)

Multimeter Tier

- ▶ Basic (Vichy DT830, ₹300): voltage + current + continuity — for hobby
- ▶ Auto-range (Mastech MS8268, ₹2k): better range handling, capacitance
- ▶ Bench DMM (Brymen BM235, ₹6k): 4-digit, fast continuity, durable
- ▶ Lab DMM (Keysight 34461A, ₹40k+): 6.5-digit, GPIB/USB, used in production test
- ▶ Clamp meter for AC currents — non-contact
- ▶ Insulation tester (megger) for safety isolation checks

Logic Analyzer

- ▶ Captures + decodes digital protocols (UART, SPI, I²C, CAN, USB)
- ▶ Way easier than using scope channels for digital protocol debug
- ▶ Saleae Logic 8 (₹15k) — 8 channels, USB, polished software
- ▶ Saleae Logic Pro 16 (₹60k) — 16 channels, 500 MHz analog mode
- ▶ Kingst LA1010 (₹3k) — budget alternative
- ▶ Built into many MSOs (mixed-signal scopes)
- ▶ Sigrok / PulseView (open source) → use cheap clones
- ▶ Auto-decode I²C / SPI / UART payloads → text output, no manual decoding

Power-Related Tools

Bench power supply

Current-limited 0-30V
+/-5A typical

Electronic load

Sink current for testing
power supplies

Current probe

Scope accessory for
A-range measurement
(Hall-effect)

Power analyzer

Measures W, PF, harmonic
for AC power testing

ATX PSU tester

Check PC power supplies

USB-C tester

Measures V, I, W
for USB-PD validation

RF / Wireless Tools

- ▶ Spectrum analyzer: measures RF signal vs frequency
- ▶ Hobby (Rigol DSA815, ₹1.5L) → 1.5 GHz coverage
- ▶ RTL-SDR (₹3k): USB dongle, software-defined, 30 MHz-1.7 GHz
- ▶ NanoVNA (₹3-8k): pocket vector network analyzer, 50 kHz-6 GHz
- ▶ Anechoic chamber: for OTA (over-the-air) testing — usually rented
- ▶ Signal generator: produces clean RF for receiver testing
- ▶ Pre-compliance EMC kit: near-field probes + low-cost receiver

Thermal Tools

- ▶ Thermal camera (FLIR ONE Pro, Seek Compact Pro — ₹15-50k)
- ▶ Plugs into smartphone — see hot spots in real time
- ▶ Industrial Flir E5 (₹2 L+): higher resolution, standalone
- ▶ Thermocouple + DMM: spot temperature measurement at specific point
- ▶ IR thermometer: contact-free spot reading
- ▶ Use cases: find hot ICs, thermal interface debugging, verify cooling solutions
- ▶ Capture before/after photos to document thermal improvements

Probing Accessories

- ▶ Standard scope probes — 10:1, 100 MHz BW
- ▶ Active differential probe (€30-100k): for differential signals, high-Z circuits
- ▶ Current probe (Hall): measures DC + AC currents non-invasively
- ▶ Pomona micro grabbers: tiny clip-on test points
- ▶ Tweezer probes: for fine-pitch SMD
- ▶ Test point clips (SOIC clips for in-circuit IC reading)
- ▶ BNC-to-grabber cables for quick probe-to-test-point connections
- ▶ Helping hands + magnifier for tiny probe placement

Building a Hobby Bench (₹25-50k)

- ▶ Rigol DS1054Z scope (₹35k) — best value 4-channel
- ▶ Saleae Logic 8 clone (₹3k) — for digital debug
- ▶ Bench supply (Owon SPE3103, ₹6k) — single-channel 30V/3A
- ▶ Mastech MS8268 multimeter (₹2k)
- ▶ Pinecil soldering iron (₹3k)
- ▶ USB microscope (₹2k)
- ▶ Total: ~₹50k — covers 90% of PCB debug needs
- ▶ Add later: thermal camera, NanoVNA, electronic load

Pro Bench Setup (₹1.5-3 L)

- ▶ Keysight DSOX2024A scope (₹70k) — 200 MHz, 4-channel + protocol decode
- ▶ Saleae Logic Pro 16 (₹60k) — 16-channel + 500 MHz analog mode
- ▶ Keysight 34461A bench DMM (₹40k) — 6.5-digit precision
- ▶ Rigol DP832 bench supply (₹35k) — 3-channel programmable
- ▶ Hot-air station 858D (₹3k)
- ▶ Pinecil + tip set (₹4k)
- ▶ FLIR E5 thermal (₹2 L) — separate purchase
- ▶ Total: ~₹2 L — professional-grade

Bench Best Practices

- ▶ ESD-safe surface + wrist strap connected to ground
- ▶ Lab notebook open — every measurement gets recorded
- ▶ Label everything — test points, jumper wires, board revs
- ▶ Photo of each setup before changing — easy to re-create
- ▶ Calibrate scope monthly (auto-cal feature on most modern models)
- ▶ Keep cables organized — tangle = wasted time + accidental shorts
- ▶ Backup the bench supply on UPS — prevents brownout damage
- ▶ Daily routine: power cycle scope + DMM to find drift

Key Takeaways

Key takeaways from this lecture

Scope = #1 tool

100 MHz minimum.

Logic analyzer = digital debug

Saleae Logic 8 entry-level.

Bench supply current-limited

Save boards from shorts.

Thermal camera = priceless

See hot spots in real time.

Multiple probes

Standard + differential + current.

ESD-safe workspace

Industry baseline.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ EEVblog tool reviews
- ▶ Marco Reps YouTube — bench builds
- ▶ Keysight Test University — free training
- ▶ Rigol / Siglent / Tektronix product pages
- ▶ Local distributors — Element14, Mouser, DigiKey
- ▶ Indian sources: Robu, Quartz Components, SP Road
- ▶ Hackaday — bench setup blog posts
- ▶ Subreddit /r/AskElectronics for tool recommendations

Thank you • Build something this week.